

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Lab 8.1 D3 Victoria LGA Map</title>
6   <script src="https://d3js.org/d3.v7.min.js"></script>
7   <style>
8     .lga {
9       fill: lightblue;
10      stroke: white;
11      stroke-width: 0.5;
12    }
13    .lga:hover {
14      fill: orange;
15    }
16  </style>
17 </head>
18 <body>
19   <h2>Lab 8.1 - Victoria LGAs (JSON)</h2>
20   <svg width="600" height="600"></svg>
21
22   <script>
23     const width = 600;
24     const height = 600;
25
26     const svg = d3.select("svg");
27
28     // Mercator projection centered on Victoria
29     const projection = d3.geoMercator()
30       .center([144.9631, -37.8136]) // Melbourne coords
31       .translate([width / 2, height / 2])
32       .scale(5000); // adjust zoom
33
34     const path = d3.geoPath().projection(projection);
35
36     // Load JSON file
37     d3.json("LGA_VIC.json").then(function(geoData) {
38       svg.selectAll("path")
39         .data(geoData.features)
40         .enter()
41         .append("path")
42         .attr("class", "lga")
43         .attr("d", path)
44         .append("title")
45         .text(d => d.properties.LGA_name || "Unknown");
46     });
47   </script>
48 </body>
49 </html>
```